Sahaj Bhadja

(213) 284-4651 sahaj.bhadja@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years of experience building **C++/Python systems**, **distributed applications**, and performance-focused **Linux tooling** with SQL, JavaScript, and React.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **concurrent backend orchestration** with explicit job states, failure isolation, recoverable processing, and observable workflows under production load reliably daily.
- Reduced dashboard latency from 1.2s to 350ms by optimizing indexed queries and **multi-tenant SQL models** supporting 500+ users with normalized schemas, row-level isolation, and efficient access patterns.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive workflows during concurrent model execution and production traffic reliably at scale.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Improved long-horizon trajectory planning by 18% in simulation by building an **algorithmic Python pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Raised closed-loop stability 15% through **quantitative performance analysis** of HD maps, ego paths, nearby agents, attention, and distance-aware interactions across diverse autonomous-driving simulations and edge cases reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows at scale.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **real-time microservice platform** using React, JavaScript, Node.js, WebSockets, REST APIs, and Azure SQL across distributed business workflows and customer-facing operational requirements.
- Reduced resource wastage by 42% through **data synchronization and testing**, integrating automated suites, Docker/Jenkins delivery, Azure Monitor, Sentry, and Kibana across modular production services reliably at scale.

Skills

Languages: C++, Python, SQL, JavaScript, React, Java, TypeScript
Systems & Algorithms: Linux, Systems Programming, Data Structures, Algorithms, OOP, Multithreading, Process Scheduling, Performance Tuning, Memory Management, Debugging, Simulation
Backend & Data: Distributed Systems, Microservices, FastAPI, Flask, REST APIs, API Design, Data Modeling, PostgreSQL, Azure SQL, Redis, WebSockets
Cloud & Engineering: AWS, Azure, AWS RDS, AWS S3, AWS Glue, Docker, Jenkins, GitHub Actions, CI/CD, Git, PyTest, Unit Testing

Projects

Low-Latency Order Book Simulator - Design Concept

- Proposed a **C++ limit-order-book concept** with price-time priority, insert, cancel, match, deterministic events, efficient data structures, replayable inputs, latency measurement, and structured debugging workflows for analysis and testing.

Market Data Replay Engine - Design Concept

- Proposed a **Python/C++ replay service** parsing timestamped market events, preserving deterministic ordering, controlling playback speed, persisting derived metrics in SQL, and supporting reproducible strategy and systems testing.

CPU Scheduling Simulator | Live

- Implemented **C++ scheduling algorithms** for FCFS, SJF, SRT, Round Robin, priority, and MLQ, with edge-case handling, process-state visualization, and dynamic turnaround and waiting-time calculations across simulations reliably.